DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA16 Data Warehousing and Data Mining

Assignment 2 – CO2

CO2 Coverage: *Data mining from a database perspective, data mining functionalities, steps in the data mining process (KDD), architecture and classification of data mining systems, basic data mining tasks and issues, data dispersion, data cleaning, integration, transformation, reduction and discretization.*

Course Outcome: CO2 – Analyse data dispersion techniques and list the various algorithms available for mining the data.

ANSWERS:

Q1 – Matching Business Problems To Mining Functionalities:

TeleNova is a telecom company with 8 million subscribers. It needs different data mining techniques to solve its six business problems.

1. P1 – Find plans and add-on services customers purchase together:

Data Mining Functionality: Association Analysis.

Learning Type: Unsupervised learning.

Knowledge Produced: Association rules that identify products or services frequently purchased together.

Required Data: Subscriber ID, plan name, add-on services and purchase date.

Algorithms:

- Apriori Algorithm: Identifies frequent itemsets and generates association rules using support and confidence.
- FP-Growth Algorithm: Finds frequent patterns efficiently without generating many candidate itemsets.

Reason:

These algorithms help TeleNova identify which plans and add-on services customers commonly purchase together.

2. P2 – Predict whether a customer will leave the network:

Data Mining Functionality: Classification and Prediction.

Learning Type: Supervised learning.

Knowledge Produced: A prediction indicating whether a customer is likely to leave the network.

Required Data: Customer age, usage, recharge history, complaints, subscription duration and previous churn status.

Algorithms:

- Decision Tree: Creates understandable rules to predict customer churn.
- Naive Bayes: Uses probability to classify customers into churn and non-churn categories.

Reason: These algorithms help TeleNova identify customers who may leave and take preventive action.

3. P3 – Divide subscribers into natural usage segments:

Data Mining Functionality: Clustering.

Learning Type: Unsupervised learning.

Knowledge Produced: Groups of customers with similar usage patterns.

Required Data: Data usage, voice calls, recharge amount, roaming and subscription details.

Algorithms:

- K-Means: Groups customers based on similarity.
- Hierarchical Clustering: Creates groups in a hierarchical structure based on customer similarities.

Reason: These algorithms help TeleNova identify customer groups such as high-data users, low-data users and frequent callers.

4. P4 – Identify SIM cards with unusual calling behaviour:

Data Mining Functionality: Outlier Analysis.
Learning Type: Unsupervised learning.
Knowledge Produced: Unusual records or SIM cards that behave differently from normal users.
Required Data: Call frequency, call duration, destination numbers, location and calling time.
Algorithms:

- Isolation Forest: Identifies unusual records by isolating them from normal data.
- Local Outlier Factor (LOF): Identifies records that are unusual compared with nearby records.

Reason: These algorithms help identify suspicious SIM cards and unusual calling patterns.

5. P5 – Compare high-value and low-value customers:

Data Mining Functionality: Concept/Class Description and Discrimination.
Learning Type: Descriptive analysis using known customer categories.
Knowledge Produced: General characteristics of high-value and low-value customers.
Required Data: Customer revenue, recharge amount, subscription plan, usage and customer lifetime.
Algorithms:

- Decision Tree: Identifies important characteristics that distinguish customer groups.
- OLAP Analysis: Compares customer information across different categories.

Reason: These techniques help TeleNova understand valuable customers and develop suitable marketing strategies.

6. P6 – Study monthly data usage over 24 months:

Data Mining Functionality: Evolution Analysis.
Learning Type: Time-oriented descriptive analysis.
Knowledge Produced: Trends, seasonal patterns and changes in customer data usage.
Required Data: Monthly data usage, subscriber ID, month, year and subscription plan.
Algorithms:

- Time-Series Analysis: Identifies trends and seasonal changes in data usage.
- Linear Regression: Estimates the direction and rate of changes in usage.

Reason: These techniques help TeleNova understand customer usage trends and plan future network requirements.

Evaluation of the all-purpose algorithm proposal:
The proposal to purchase one all-purpose algorithm is not suitable because each business problem requires a different data mining functionality.

Association analysis identifies relationships between purchased services, classification predicts customer churn, clustering groups similar subscribers, and outlier analysis detects unusual behaviour. Evolution analysis identifies changes over time.

Therefore, TeleNova should use a combination of data mining algorithms instead of depending on a single algorithm.

Q2 – The KDD Process, System Architecture and Mining Issues:

MediCore is a hospital network that wants to identify factors responsible for patient readmission within 30 days.

A. KDD Process for MediCore:
KDD stands for Knowledge Discovery in Databases. It is a process used to discover useful knowledge from large amounts of data.

1. Data Selection

MediCore collects relevant patient information from admission, laboratory, pharmacy and billing databases. It selects patient records that are useful for predicting readmission.

2. Data Preprocessing

The collected data is cleaned by removing duplicate records, correcting incorrect values and handling missing information. Patient identifiers and dates are also checked for consistency.

3. Data Transformation

The cleaned data is transformed into a suitable format for mining. New attributes such as patient age, hospital stay duration, previous admissions and medication count are created.

4. Data Mining

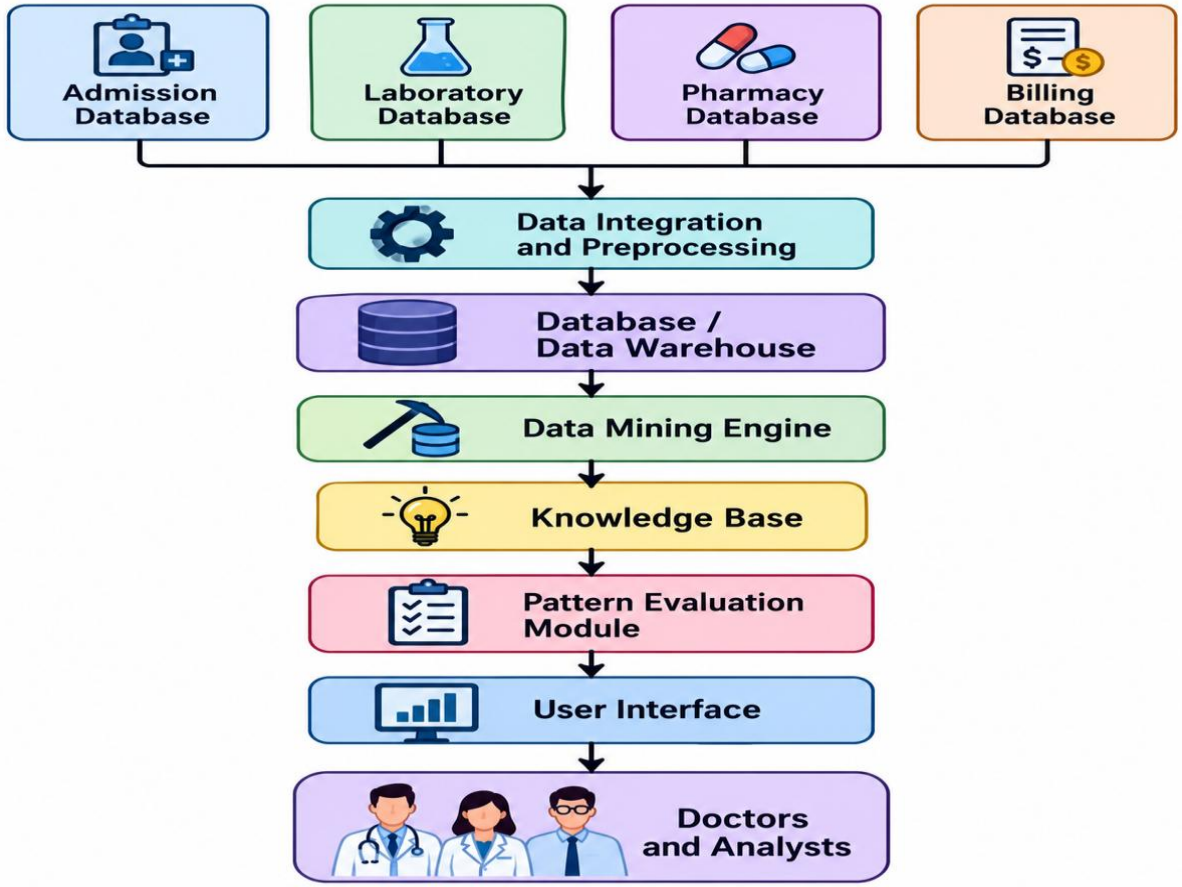
Classification algorithms such as Decision Tree, Naive Bayes and Random Forest are applied to identify patients who may be readmitted within 30 days.

5. Interpretation and Evaluation

The results are evaluated using accuracy, precision, recall and F1-score. Doctors and hospital administrators examine the discovered patterns and use them to improve patient care.

B. Architecture of a Data Mining System:

The architecture of a typical data mining system contains the following components:



Components:

1. Database or Data Warehouse Server: Stores integrated patient information from different hospital systems.
2. Knowledge Base: Stores medical knowledge, rules, metadata and previously discovered patterns.
3. Data Mining Engine: Applies mining algorithms to identify useful patterns and predict patient readmission.

4. Pattern Evaluation Module: Evaluates the discovered patterns and identifies useful results.
5. User Interface: Allows doctors and analysts to interact with the system and view reports.

C. Classification of the Data Mining System:

1. Based on Kinds of Data

The system uses relational databases, patient records, laboratory results, pharmacy transactions and billing information.

2. Based on Kinds of Knowledge

The system discovers classification rules, prediction results, patient profiles, relationships and patterns.

3. Based on Techniques Used

It uses machine learning, statistical analysis, classification, prediction and data visualization techniques.

4. Based on Applications

It is used in healthcare analytics, patient readmission prediction, hospital resource management and improving patient care**.**

D. Why SQL and Basic Statistics Are Not Enough:

SQL is used to retrieve and manage data from databases. Basic statistics is used to calculate values such as averages, percentages and totals.

Data mining goes beyond these operations by discovering hidden patterns and predicting future outcomes.

For example, SQL can retrieve the list of patients who were readmitted within 30 days. Statistics can calculate the average readmission rate. However, a data mining model can identify combinations of factors such as previous admissions, long hospital stays and abnormal laboratory results that are associated with readmission.

Therefore, data mining provides deeper knowledge and predictive information than ordinary database queries and basic statistical summaries

Q3 – Describing and Comparing Data Dispersion:

A. Mean, Median, Mode and Midrange:

Given 24 delivery times:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95.

Mean = 1143 / 24 = 47.625

Median = (43 + 45) / 2 = 44

Mode = 35 and 42

Midrange = (28 + 95) / 2 = 61.5

The mean is greater than the median because the extreme value 95 increases the average. This indicates positive skewness.

B. Dispersion Measures:

Using the median-of-halves quartile method:

- Range = 95 − 28 = 67
- Q1 = 37
- Q3 = 54
- IQR = 54 − 37 = 17
- Sample variance = 222.59
- Standard deviation = 14.92

C. Five-Number Summary and Outlier:

Five-number summary:

- Minimum = 28
- Q1 = 37
- Median = 44
- Q3 = 54
- Maximum = 95

Outlier limits:

- Lower limit = 37 − (1.5 × 17) = 11.5
- Upper limit = 54 + (1.5 × 17) = 79.5

Since 95 is greater than 79.5, 95 is an outlier.

D. After Removing 95:

Measure	Original	After Removal
Mean	47.625	45.57
Median	44	43
Mode	35, 42	35, 42
Midrange	61.5	50
Range	67	44
IQR	17	17
Variance	222.59	126.26
Standard deviation	14.92	11.24

The range, variance and standard deviation change considerably because they are affected by extreme values.

E. Classification of Measures:

Measure	Type
Mean	Algebraic
Median	Holistic
Variance	Algebraic
Standard deviation	Algebraic
Range	Distributive

F. Verification and Recommendation:

Python, Excel or KNIME can verify the calculations. Differences may occur because of different quartile methods or population and sample variance formulas.

The median of 44 minutes is recommended as the typical delivery time because it is less affected by outliers. The IQR should also be reported to show variation.

Q4 – Cleaning Noisy and Incomplete Data:

A. Identify Data Defects:

Meter	Defect	Type	Solution
M102	Missing kWh	Missing value	Use suitable imputation.
M103	north instead of North	Inconsistency	Standardize the region name.
M104	−14 kWh	Invalid value	Verify and mark invalid if incorrect.
M105	Missing household type	Missing value	Infer or mark Unknown.
M105	2300 V	Invalid value	Verify the reading and correct only if confirmed.
M106	S. instead of South	Inconsistency	Standardize the region name.

B. Equal-Frequency Binning:

Given values:

5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38.

There are 12 values and 4 bins. Each bin contains 3 values**.**

Bin	Values	Mean
1	5,9,11	8.33
2	14,18,19	17
3	23,26,27	25.33
4	30,33,38	33.67

Smoothing by bin means:

- Bin 1: 8.33, 8.33, 8.33
- Bin 2: 17, 17, 17
- Bin 3: 25.33, 25.33, 25.33
- Bin 4: 33.67, 33.67, 33.67

Smoothing by bin boundaries:

- Bin 1: 5, 11, 11
- Bin 2: 14, 19, 19
- Bin 3: 23, 27, 27
- Bin 4: 30, 30, 38

C. Regression and Clustering:

Regression predicts expected readings using other attributes. Large differences between actual and predicted values can indicate noise.

Clustering groups similar readings and identifies unusual values that fall far from normal groups.

A reading of 2300 V is suspicious and must be verified, while 5 kWh is a valid low consumption value.

D. Effect of Global Mean Imputation:

Replacing missing values with the global mean reduces data dispersion and may reduce variance. It can also hide important differences between groups and affect data mining results. Therefore, class-wise mean, median or regression-based imputation may be more suitable depending on the data

Q5 – Integration and Transformation of Data:

Part A – Integration

A. Identify Conflicts

Conflict	Example	Solution
Entity identification	student_id and sid	Use a common student ID.
Naming	name and full_name	Use student_name.
Format	Different DOB formats	Convert into one standard format.
Value	scholarship and scholarship_flag	Convert into Yes/No values.

B. Redundant Attributes:

Data integration may create duplicate attributes because the same information is stored in different databases. Correlation analysis helps identify highly related attributes and possible redundancy.

Part B – Redundancy Analysis:

A. Pearson Correlation:

Attendance = 60, 70, 75, 80, 85, 90.

Marks = 52, 58, 66, 70, 79, 85.

Using the Pearson correlation formula:

$$r = \frac{\sum(x - \bar{x})\,(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Answer: r = 0.9851

This indicates a very strong positive relationship between attendance and marks

B. Chi-Square Test:

Expected frequency formula:

$$E = \frac{Row\ Total \times Column\ Total}{Grand\ Total}$$

Expected values:

	Scholarship	**No Scholarship**
Hostel	85.71	214.29
Day scholar	114.29	285.71

Chi-square formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Calculated value:

$$\chi^2 = 33.60$$

Critical value = 3.84.

Since 33.60 is greater than 3.84, hostel status and scholarship are not independent at the 0.05 significance level**.**

Part C – Transformation:

A. Normalization

Given incomes: ₹20,000, ₹35,000, ₹50,000, ₹80,000 and ₹1,50,000.

Income	Min-Max	Z-Score	Decimal Scaling
20,000	0.0000	-1.0217	0.020
35,000	0.1154	-0.6957	0.035
50,000	0.2308	-0.3696	0.050
80,000	0.4615	0.2826	0.080
1,50,000	1.0000	1.8043	0.150

B. Euclidean Distance

Students:

- Student 1 = (19, 35,000)
- Student 2 = (21, 80,000)

Before normalization:

$$D = \sqrt{(21 - 19)^2 + (80000 - 35000)^2}$$

Distance ≈ 45,000

After min-max normalization:

- Student 1 = (0.25, 0.11538)
- Student 2 = (0.50, 0.46154)

$$D = \sqrt{(0.50 - 0.25)^2 + (0.46154 - 0.11538)^2}$$

Distance ≈ 0.4270

Normalization prevents income from dominating the distance calculation because of its larger numerical scale.

C. Transformation Examples:

1. Aggregation: Calculate monthly attendance from daily attendance records.
2. Generalisation: Convert date of birth into age groups.
3. Attribute Construction: Create an academic performance category using marks and attendance.

Q6 – Data Reduction and Discretization:

Part A – Data Reduction:

A. Reduction Techniques:

1. Data-Cube Aggregation: Combines detailed sensor readings into daily or monthly summaries.
2. Attribute Subset Selection: Removes irrelevant sensor attributes using forward selection, backward elimination or decision trees.
3. PCA: Reduces the number of attributes by creating principal components from correlated variables.
4. Numerosity Reduction: Uses sampling, histograms and clustering to reduce data volume.
5. Compression: Reduces storage requirements while preserving useful information.

B. Sampling Methods

Method	Description
Without Replacement	A record is selected only once.
With Replacement	A record can be selected multiple times.
Cluster Sampling	Selects complete groups of records.
Stratified Sampling	Selects records from different categories.

Best Method: Stratified Sampling

Since only 0.5% of the records represent failures, stratified sampling helps maintain the representation of failure records. Class weighting or oversampling can also be used during training.

C. Principal Components:

PCA components can be selected using cumulative explained variance. A common starting point is to retain components that explain at least 95% of the total variance. Failure detection performance must also be checked.

Part B – Discretization:

A. Equal-Width Binning:

Given:

3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44.

Number of intervals = 4.

$$Width = \frac{44 - 3}{4} = 10.25$$

Interval	Values	Count
3–13.25	3, 7, 7, 9, 12	5
13.25–23.5	14, 14, 15, 18, 21, 21, 22	7
23.5–33.75	25, 25, 27, 31	4
33.75–44	36, 36, 40, 44	4

B. Equal-Frequency Binning:

Each bin contains 5 values.

Bin	Values	Count
1	3, 7, 7, 9, 12	5
2	14, 14, 15, 18, 21	5
3	21, 22, 25, 25, 27	5
4	31, 36, 36, 40, 44	5

C. Entropy-Based Split:

Parent entropy = 1 bit.

Split Point	Information Gain
11	0.1080
13.5	0.2365
16.5	0.3958
20	0.6100
23.5	0.2781
26.5	0.1245
29.5	0.3958
33	0.2365
37.5	0.1080

The highest information gain is 0.6100 at split point 20. Therefore, 20 is the best split point.

D. Comparison:

Equal-width and equal-frequency binning are unsupervised techniques because they do not use class labels.

Entropy-based discretization is supervised because it uses class labels to find the best split.

Supervised discretization is generally more suitable for classification because it creates intervals that help separate different classes**.**

Q7 – Diagnose, Redesign and Validate:

A. Diagnose the Original Pipeline:

Problem	Explanation
Global mean imputation	Reduces variation and may distort income distribution.
Occupation removed	Important information may be lost.
Equal-width discretization	Unsuitable for highly skewed income.
Normalization before splitting	Causes data leakage.
Random 10% training sample	May not contain enough useful default records.
Only 3% defaults	Creates class imbalance.
97% accuracy	Does not prove that the model predicts defaulters correctly.

A model that predicts every customer as a non-defaulter can achieve approximately 97% accuracy but will miss all defaulters.

B. Design Objectives:

The main objective is to improve default prediction.

1. Increase default recall.
2. Improve precision and F1-score.
3. Reduce false negatives.
4. Prevent data leakage.
5. Preserve important attributes.

An initial target could be at least 70% recall for the default class, with an acceptable precision and investigation workload.

C. Revised Preprocessing Strategy:

1. Split the dataset into training and testing sets before preprocessing.
2. Remove duplicate and incorrect records.
3. Replace missing income values using the training-set median.
4. Retain occupation and replace missing values with Unknown or a suitable training-set value.
5. Apply log transformation to reduce income skewness.
6. Use class weighting or SMOTE on training data to handle class imbalance.
7. Select important attributes and remove irrelevant features.
8. Train Decision Tree, Random Forest or Logistic Regression models.
9. Select the prediction threshold using validation data.
10. Evaluate using accuracy, precision, recall, F1-score and confusion matrix.

All preprocessing parameters must be learned from training data only.

D. Prediction Before Implementation:

The redesigned pipeline is expected to reduce the influence of extreme income values and improve the representation of defaulters during training.

Accuracy may decrease slightly, but default recall is expected to increase because the model will identify more defaulters.

E. Implementation and Validation:

Use a public dataset such as UCI German Credit and implement the original and redesigned pipelines in Python, Weka or KNIME.

The following metrics should be compared:

Metric	Original Model	Redesigned Model
Accuracy	Record actual result	Record actual result
Precision	Record actual result	Record actual result
Recall	Record actual result	Record actual result
F1-score	Record actual result	Record actual resul

The results must be filled in after running both models on the same test data. They should not be estimated or invented.

F. Defend the Final Design:

The redesigned pipeline is better because it handles missing values, retains useful attributes, prevents data leakage and addresses class imbalance.

It focuses on improving the identification of actual defaulters rather than depending only on accuracy.

One remaining risk is that future customer behaviour may differ from the training dataset. Therefore, the bank should regularly monitor model performance and update the model when necessary.